What Does a Tuned System's Benchmark Score Mean? Optimism After Prompt and Program Search, and How to Report It

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Abstract

A tuned LLM system is usually reported with its score on the same benchmark items that were used to search for its prompt or program. We ask how far such a score is from performance on fresh items, what drives the gap, whether it changes conclusions, and how to report more reliably. In a synthetic world with exact truth (203,400 search replicates) and on a real model and benchmark (Qwen2.5-1.5B-Instruct on MMLU-Pro, 300 candidate configurations replayed from cached outputs), the naive report is optimistic by up to 14.6 points (synthetic, 50 items, 200 candidates, no true differences) and 5.7 points on MMLU-Pro (100 search items, 300 candidates), roughly doubling the apparent gain over the baseline; the gap shrinks like $1/\sqrt{n}$, grows with the number of candidates and few-shot demonstrations drawn from the search set, and follows an extreme-value formula whose parameters we fit to each setting (a check of its form, not a held-out prediction). Naive reports declare a tuned system better than an untuned one in 97.6% of synthetic replicates with no true gain; on MMLU-Pro such false-claim rates are not identifiable (the baseline is almost always truly worse), so decision errors rest on the synthetic world and a paired A-versus-B design on real outcomes, where reports rank systems by search effort. Hold-out and conditional/hybrid selection-aware intervals restore nominal coverage but cost items, width and power; empirical-Bayes disclosure fails. Results rest on one small model, template and ensemble configurations replayed from cached outputs on one fixed 2000-item pool (between-pool uncertainty is not estimated), and no live tool run.

1 Introduction

Automatic prompt and program optimisation has become routine: a tool such as DSPy [Khattab et al., 2023] or MIPRO [Opsahl-Ong et al., 2024] searches instructions, demonstrations or multi-call pipelines on a set of benchmark items and the best configuration is reported with its score on benchmark items. When those items are the *same* ones that drove the search, the number is a maximum over many noisy scores, not an unbiased estimate: it measures the winner's luck as well as the winner's quality. The consequence for the reader is unclear, because the size of the optimism, what governs it, and whether the comparisons built on such numbers ("our tuned system beats the baseline", "A beats B") survive it are rarely examined for LLM systems.

We study this as a statistical question in a controlled setting. A search procedure with a budget evaluates K configurations on n items, returns $\hat{c}$, and the *naive report* is the accuracy of $\hat{c}$ on those n items; the target is its accuracy on fresh items. We ask (Q1) how large the gap is, what drives it, and whether it changes conclusions; and (Q2) whether more reliable reporting methods exist and how to test that they are.

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Contributions. (1) A measurement of the optimism of the naive report in a synthetic world with exact truth (216 cells, 203,400 replicates) and on a real LLM and benchmark (Qwen2.5-1.5B-Instruct on 2000 MMLU-Pro questions, 300 prompt/ensemble configurations, 52 replay cells of 1000 replicates), with an extreme-value formula whose parameters are fitted in each setting (it checks the functional form; it is not a held-out prediction). (2) The decision-level consequence: naive reports declare a tuned system better than an untuned one when it is not, and rank two tuned systems by search effort and search-set size rather than by quality. These error rates rest on the synthetic world and, for real outcomes, on a paired A-versus-B design; the "tuned beats baseline" false-claim rate is not identifiable on MMLU-Pro, where the baseline is almost always truly worse. (3) A validation protocol with known truth and a real outcome matrix, applied to the reporting methods M0–M5 (naive, hold-out, cross-fitting, bootstrap optimism correction, simultaneous and conditional/hybrid selection-aware intervals, empirical-Bayes and Ladder-style disclosure), with an honest account of where each fails. (4) Scope: one small model on CPU, a prompt-format family rather than a search-heavy pipeline, emulated rather than live search (no DSPy/MIPRO run; the live tier and the 0.5B arm were not run), and a single fixed item pool whose between-pool variability is not estimated (Sec. 5); our real-LLM claims are limited accordingly.

2 Related Work

Prompt and program optimisation. DSPy compiles declarative LM programs by searching instructions and demonstrations against a metric on a training set [Khattab et al., 2023]; MIPRO jointly optimises instructions and demonstrations of multi-stage programs [Opsahl-Ong et al., 2024]; best-arm-identification formulations select prompts under a limited evaluation budget [Shi et al., 2024]. These works report the score of the selected configuration on held-out data of a given size; we ask what such a number means when the same items drive selection and scoring, and how it depends on the search budget.

Evaluation variability of prompts. Prompt format alone changes accuracy substantially [Sclar et al., 2024], and prompt-performance distributions can be estimated from few evaluations [Polo et al., 2024]. This supplies the between-configuration spread that, in our analysis, drives the selection effect.

Adaptive data analysis and test-set reuse. Reusing a holdout adaptively invalidates naive estimates [Dwork et al., 2015]; the Ladder reports thresholded leaderboard scores to limit overfitting [Blum and Hardt, 2015]. Empirical studies of test-set reuse find smaller drift than worst-case bounds suggest [Recht et al., 2018, Mania et al., 2019]. We study the same phenomenon for automatic prompt/program search with an explicit selection step, and compare the Ladder-style report as a baseline.

Selection-aware inference. Winner's-curse corrections and conditional/hybrid intervals for selected targets [Andrews et al., 2024], simultaneous inference over selected models [Berk et al., 2013], Gaussian multiplier-bootstrap bounds for maxima [Chernozhukov et al., 2013], and the target-of-estimation analysis of cross-validation [Bates et al., 2024] provide the tools we adapt to reporting a tuned system's score.

3 Setting, Estimands and Reporting Methods

Setting. A system has a configuration space $\mathcal{C}$ (an instruction/prompt, or a multi-call program with demonstrations). A search procedure S with budget B evaluates configurations on a *search set* D of n benchmark items and returns a configuration $\hat{c}$. Let $\theta(c)$ be the accuracy of c on the item population (fresh items) and $R(c; D)$ its accuracy on D. The *naive report* is $R(\hat{c}; D)$.

Estimands. The optimism is $\Delta = R(\hat{c}; D) - \theta(\hat{c})$; we study $\mathbb{E}[\Delta]$, its quantiles, and the bias of the naive report for $\theta(\hat{c})$. Reporting methods are judged on bias, RMSE, coverage of $\theta(\hat{c})$ at nominal 95%, interval width, data cost, and the error rates of claims made from them ("tuned beats baseline by margin m", "A beats B").

Reporting methods. M0 naive; M1 holdout (search on D_s, report on disjoint D_r); M2 cross-fitting (repeat search per fold, pooled out-of-fold score; estimates the expected fresh accuracy of the *procedure*, not of one $\hat{c}$); M3 bootstrap optimism correction; M4 selection-aware inference (simultaneous multiplier-bootstrap bound over all evaluated configurations; winner's-curse-adjusted

intervals); M5 disclosure of the searched-candidate score distribution with K and n, and a Ladder-style thresholded report.

Extreme-value approximation. For exhaustive search over K candidates with per-item outcome SD σ (candidate accuracy θ_0, $\sigma^2 = \theta_0(1-\theta_0)$), binary item-outcome correlation r between candidates and true-accuracy spread τ_p (probability scale), $s^2 = \sigma^2(1-r)/n$ is the candidate-specific part of the sampling variance of a search-set score, and we use

$$\mathbb{E}[\Delta] \;\approx\; \frac{s^2\, g(K)}{\sqrt{\tau_p^2 + s^2}}, \qquad g(K) = \mathbb{E}\max_{k \le K} Z_k,\; Z_k \overset{iid}{\sim} \mathcal{N}(0,1), \tag{1}$$

the winner's-curse of the maximum of K noisy scores around a true spread τ_p; at $\tau_p = 0$ it reduces to $s\, g(K)$. It is a heuristic (Gaussian, independent-noise approximation of correlated binary scores), checked against simulation and the real matrix rather than derived with error bounds.

4 Experiments

Every number below is read from files in `results/` produced by the shipped code (script map in `figures/SCRIPTS.md`; commands in `REPRODUCE.md`). Intervals over replicates are mean$\pm$1.96 SE; coverage and decision rates carry Wilson 95% intervals (T0); on the real matrix (T1) replicates share one item pool, so T1 intervals are delete-a-block.

4.1 Synthetic world with exact truth (T0)

Design. A latent-probit world: each configuration c has a true accuracy $\theta_c = \Phi(a + \delta_c)$ with spread τ (SD of δ_c on the probit scale; $\tau = 0$ means all configurations are equal), item difficulty is shared across configurations with latent correlation ρ of configuration$\times$item effects, and outcomes are Bernoulli. Because $\theta(\hat{c})$ is known exactly, bias and coverage are measured without noise on the truth. The grid covers $n \in \{50, \ldots, 2000\}$, $K \in \{2, \ldots, 200\}$ candidates, $\tau \in \{0, .05, .1, .2\}$, ρ, baseline accuracy, three search procedures (exhaustive, successive halving, greedy coordinate edit) and few-shot demonstrations drawn from D: 216 cells and 203,400 replicates (300–1000 per cell). The reference world is $\rho = 0.6$, baseline accuracy 0.3, exhaustive search; the headline cells have 1000 replicates.

How large is the gap, and what drives it? (Fig. 1.) The naive report is optimistic in every cell with $K > 1$ configurations compared. With $\tau = 0$ (nothing to find) the mean optimism is 14.6 [14.2, 14.9] accuracy points at $n = 50$, $K = 200$, 4.5 [4.4, 4.6] at $n = 500$ and 2.2 [2.1, 2.2] at $n = 2000$; at $n = 200$ it is 2.9 [2.8, 3.1] for $K = 5$ and 5.7 [5.5, 5.8] for $K = 50$. It shrinks roughly as $1/\sqrt{n}$ and grows with K like the expected maximum of K noise terms; the closed-form extreme-value approximation $s^2 g(K)/\sqrt{\tau_p^2 + s^2}$ (Sec. 3) is within 0.84 points of the measurement over all 128 reference cells (mean theory$-$measurement -0.13 points). A larger true spread τ *reduces* the optimism (the search finds real signal), but at $K = 200$, $n = 50$ it remains 9.0 points at $\tau = 0.2$. Fig. 2: search procedures that evaluate fewer candidates on all items (halving, greedy) are less optimistic than exhaustive search (at $n = 200$, $K = 50$, $\tau = 0$: 5.0 and 4.3 vs. 5.7 points), high correlation between configurations' item outcomes reduces it (7.0 points at $\rho = 0.2$, 3.9 at $\rho = 0.9$), baseline accuracy matters little (4.6, 5.7, 6.2 points at 0.15, 0.30, 0.50), and drawing few-shot demonstrations from D adds optimism (at $K = 24$, $n = 50$, $\tau = 0$: 13.1 vs. 10.0 points; at $n = 200$: 5.7 vs. 4.9).

Does it change conclusions? (Tables 2, 7.) Take the claim "tuned beats the untuned baseline by more than $m = 2$ points" with $K = 200$, $n = 100$. When the truth is $\tau = 0$ (no gain), the naive point comparison claims it 97.6 [96.5, 98.4]% of the time. Comparing two tuned systems on reported scores, the system searched over $K = 200$ candidates is declared better than one searched over $K = 5$ in 86.7 [84.5, 88.7]% of replicates although they are equal, and a system tuned on $n = 50$ items beats one tuned on $n = 500$ in 93.2 [91.5, 94.6]% of replicates ($K = 50$). The reported ranking thus tracks search effort and set size rather than quality.

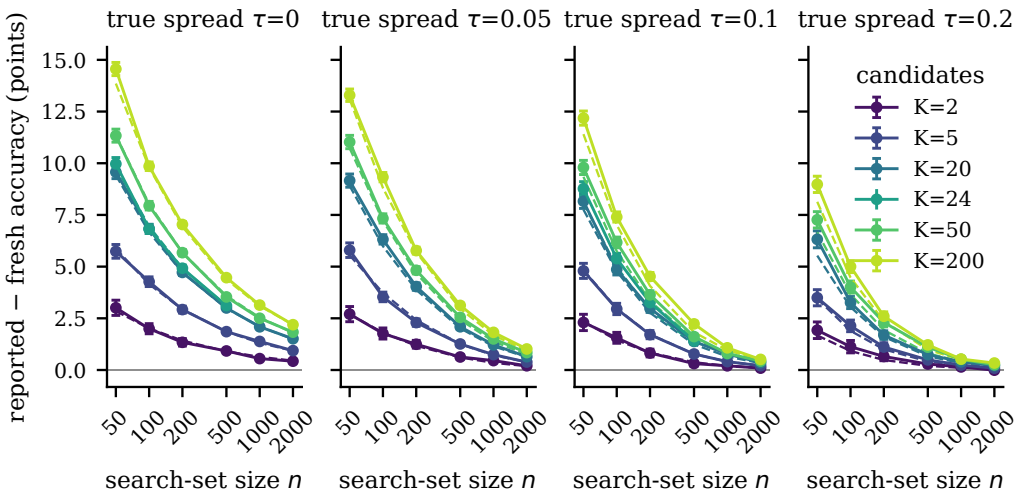


Figure 1: **Optimism of the naive report in the synthetic world (T0).** Reported minus fresh-item accuracy of the selected configuration against search-set size n, for K candidates searched exhaustively, one panel per true spread τ. Points: mean over 1000 replicates per cell, bars: 95% CI; dashed: extreme-value prediction. Reference world $\rho = 0.6$, baseline accuracy 0.3.

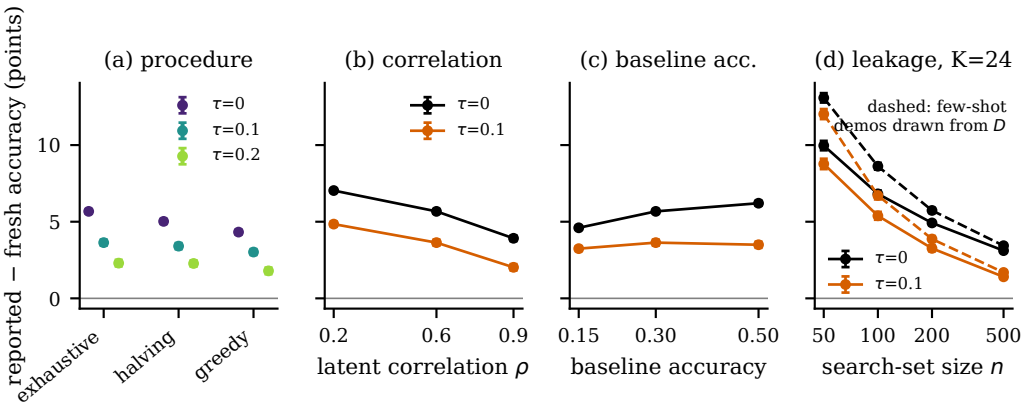


Figure 2: **Other drivers of optimism (T0).** Same quantity; $n = 200$, $K = 50$ unless stated. (a) search procedure, (b) correlation ρ between configurations' item outcomes, (c) baseline accuracy, (d) few-shot demonstrations drawn from D (dashed) or not, $K = 24$. 1000 replicates per point.

4.2 Reporting methods: calibration and decisions (T0)

Targets and metrics. Each method reports an estimate and, where available, a nominal 95% interval for $\theta(\hat{c})$ (M2: for the expected fresh accuracy of the fold-selected configurations). We measure bias, RMSE, coverage, width and the number of items consumed (Table 1, Fig. 3).

Findings (T0, $K = 200$, $\tau = 0$, $n = 100$; Table 1). The naive interval covers $\theta(\hat{c})$ in only 34.1 [31.2, 37.1]% of replicates (bias $+9.85 \pm 0.21$ points); even at $n = 500$ coverage is 41.4%, and at $\tau = 0.1$ it is 68.0% and 82.7% at $n = 100$ and 500. (i) *Hold-out* (M1) is unbiased and covers at the nominal level (0.96 for all three split fractions), but it pays in items and width: 19.7 to 37.3 points wide at $n = 100$ and, with half of D held out, RMSE 6.4 vs. 10.4 points for M0. (ii) *Cross-fitting* (M2) is unbiased for the expected fresh accuracy of the procedure's selections but its interval undercovers (0.88 [0.86, 0.90]; 0.86–0.95 over the 128 reference cells), because fold scores are dependent. (iii) *Selection-aware inference* (M4): the simultaneous bound covers (0.986) but is as biased as M0 and wide (34.2 points); the conditional (0.961) and hybrid (0.964, width 27.3 points) intervals reach

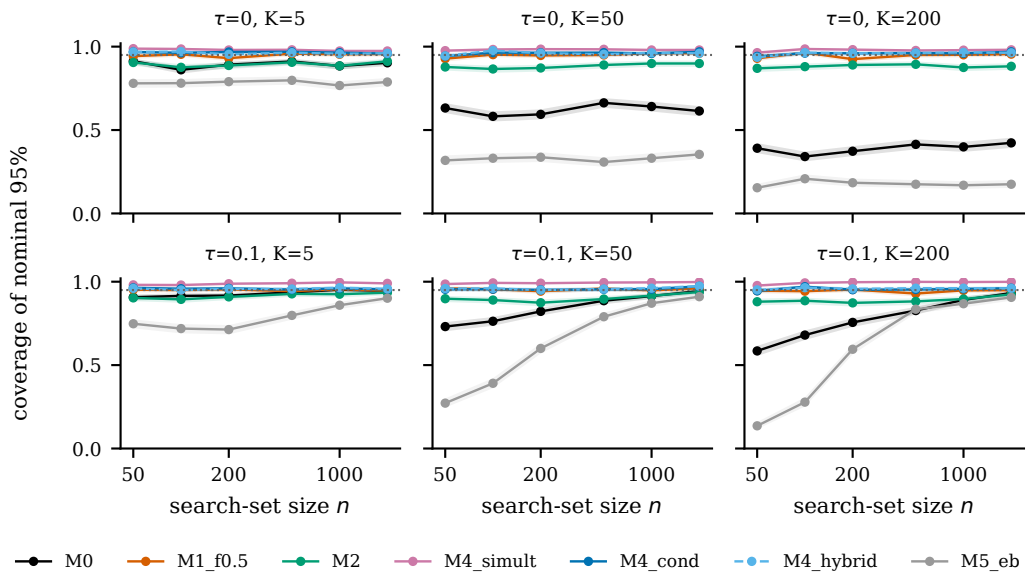


Figure 3: **Coverage of nominal-95% intervals (T0).** Wilson 95% bands over 1000 replicates; dotted: 0.95. M0: same-set interval; M1_f0.5: hold-out with half of D for selection; M2: 5-fold cross-fitting; M4_simult: simultaneous max-type bound; M4_cond, M4_hybrid: conditional/hybrid selection-aware intervals; M5_eb: empirical Bayes from disclosed scores.

Table 1: **Calibration of reporting methods (T0),** $K = 200$**,** $n = 100$**, 1000 replicates per cell** ($n = 500$ and the other methods in the appendix, Table 5). Bias and RMSE in accuracy points; coverage with Wilson 95% interval; width in points; items used for the report. Ladder/optimism-corrected point reports have no interval and appear in the appendix.

τ	n	method	bias (pts)	RMSE (pts)	coverage [Wilson 95%]	width (pts)	items for report
0	100	M0	+9.85 ± 0.21	10.40	0.341 [0.312, 0.371]	18.9	100
0	100	M1_f0.5	-0.08 ± 0.40	6.38	0.964 [0.951, 0.974]	24.6	50
0	100	M2	-0.03 ± 0.34	5.49	0.880 [0.858, 0.899]	17.7	100
0	100	M4_simult	+9.85 ± 0.21	10.40	0.986 [0.977, 0.992]	34.2	100
0	100	M4_cond	-3.99 ± 1.86	30.22	0.961 [0.947, 0.971]	45.8	100
0	100	M4_hybrid	+2.61 ± 0.46	7.94	0.964 [0.951, 0.974]	27.3	100
0	100	M5_eb	-0.09 ± 0.17	2.81	0.208 [0.184, 0.234]	1.3	100
0	100	M5_ladder	-1.17 ± 0.63	10.26	–	–	100
0.1	100	M0	+7.40 ± 0.25	8.40	0.680 [0.650, 0.708]	19.1	100
0.1	100	M1_f0.5	-0.29 ± 0.42	6.73	0.943 [0.927, 0.956]	25.4	50
0.1	100	M2	-0.35 ± 0.37	5.90	0.886 [0.865, 0.904]	18.4	100
0.1	100	M4_simult	+7.40 ± 0.25	8.40	0.993 [0.986, 0.997]	34.6	100
0.1	100	M4_cond	-3.21 ± 1.44	23.42	0.970 [0.957, 0.979]	41.5	100
0.1	100	M4_hybrid	+1.34 ± 0.46	7.52	0.964 [0.951, 0.974]	27.5	100
0.1	100	M5_eb	-4.67 ± 0.24	6.05	0.278 [0.251, 0.307]	5.3	100
0.1	100	M5_ladder	-1.48 ± 0.63	10.20	–	–	100

nominal coverage (0.94–0.98 over the 128 reference cells, as for M1_f0.5 at 0.92–0.97), but the conditional point estimate is heavy-tailed (RMSE 30 points at $n = 100$) so only its interval is usable, and hybrid has bias $+2.6$ points. (iv) Bootstrap optimism correction (M3) removes only part of the bias ($+1.9$ points for $n = 200$; $+2.5$ for the .632 variant, vs. $+7.0$ for M0; appendix). (v) The empirical-Bayes interval from the disclosed scores (M5_eb) fails: coverage 0.21 at $\tau = 0$, $K = 200$, $n = 100$, and 0.09–0.93 across the 128 reference cells (Fig. 3), since candidates share items; we report it as a negative result. The Ladder-style report has no interval; its point report is -1.2 ± 0.6 points biased with step $n^{-1/3}$ but $+8.2 \pm 0.3$ with the $0.5/\sqrt{n}$ step, i.e. its calibration depends on a free step size.

Table 2: **False-claim rate and power of "tuned beats baseline by > 2 points" (%, Wilson 95%), $K = 200$, $n = 100$ (T0).** "point": estimate difference exceeds m; "lower bound": the method's 95% lower bound exceeds the baseline's estimate $+m$. At $\tau = 0$ the true gain is 0. At $\tau = 0.1$, 162 false-claim and 608 power replicates (per-cell counts in `tables/table_d_all_cells.csv`).

method	τ=0 false claim, point	τ=0 false claim, lower bound	τ=0.1 false claim, point	τ=0.1 power, point	τ=0.1 power, lower bound
M0	97.6 [96.5, 98.4]	38.4 [35.4, 41.5]	95.0 [90.7, 97.3]	99.8 [99.1, 99.9]	73.7 [70.6, 76.6]
M1_f0.2	38.0 [35.0, 41.0]	2.2 [1.5, 3.3]	36.0 [31.4, 40.8]	77.0 [73.5, 80.1]	18.3 [15.4, 21.5]
M1_f0.5	40.1 [37.1, 43.2]	3.2 [2.3, 4.5]	42.8 [36.6, 49.2]	70.0 [66.7, 73.2]	15.4 [13.1, 18.2]
M1_f0.8	40.2 [37.2, 43.3]	2.8 [1.9, 4.0]	40.1 [33.4, 47.2]	66.1 [62.8, 69.3]	12.6 [10.5, 15.1]
M2	38.7 [35.7, 41.8]	4.1 [3.0, 5.5]	31.5 [24.8, 39.0]	74.9 [71.9, 77.8]	27.4 [24.5, 30.6]
M4_simult	97.6 [96.5, 98.4]	0.5 [0.2, 1.2]	95.0 [90.7, 97.3]	99.8 [99.1, 99.9]	17.3 [14.9, 20.0]
M4_cond	57.5 [54.4, 60.5]	1.6 [1.0, 2.6]	55.3 [48.0, 62.4]	75.6 [72.6, 78.5]	10.8 [8.9, 13.2]
M4_hybrid	57.6 [54.5, 60.6]	1.6 [1.0, 2.6]	55.9 [48.5, 62.9]	77.3 [74.4, 80.1]	16.9 [14.5, 19.6]
M5_eb	28.0 [25.3, 30.9]	23.0 [20.5, 25.7]	18.4 [13.4, 24.8]	52.4 [49.0, 55.8]	30.8 [27.8, 34.1]
M5_ladder	34.7 [31.8, 37.7]	–	48.6 [41.4, 55.9]	63.3 [60.0, 66.6]	–

Decisions (Table 2). Naive reports make a false improvement claim in 97.6% of replicates when there is no improvement; using the naive lower bound this is still 38.4 [35.4, 41.5]%. Hold-out, cross-fitting and the selection-aware lower bounds bring the false-claim rate to 0.5–4.1% with a lower-bound rule, but at this sample size the price is power: at $\tau = 0.1$ the power to detect a true gain of ≥ 2 points falls from 73.7 [70.6, 76.6]% (naive lower bound, which is not valid) to 10.8–27.4% (valid rules). The point-comparison rule has a high false-claim rate even for unbiased methods (32–43% for M1/M2) because the difference of two noisy scores exceeds $m = 2$ points by chance; *unbiasedness does not by itself license a claim*.

4.3 Real LLM on real MMLU-Pro (T1)

Design. The system is Qwen2.5-1.5B-Instruct [Qwen Team, 2024] (Q8_0 GGUF, llama.cpp, CPU; Apache-2.0) answering MMLU-Pro [Wang et al., 2024] (pinned revision, MIT license, 10 options) by reading out the log-probabilities of the ten option letters. The configuration family has 24 base prompts (4 instruction styles $\times$ 2 answer cues $\times$ {0-shot, two 3-shot demonstration sets from the validation split}) plus the 276 two-prompt logit-average ensembles of them as a pipeline-like extension: $K_{\max} = 300$ candidates. All 300 are scored once on a fixed pool of 2000 randomly drawn test questions and the per-item outcomes are cached; a search is then *replayed*: draw a search set D of n pool items, let the procedure choose $\hat{c}$ among K candidates (the baseline prompt plus $K - 1$ random ones; $K = 24$ is the base family), and score $\hat{c}$ on the remaining pool items minus a left-out block (1300–1750 fresh items, binomial SE $\approx$1.1–1.3 points, independent of the search). The pool accuracies of the 24 base prompts range from 20.5% to 29.2% and the best of all 300 is 30.1%; the between-candidate spread is small ($\tau \approx 0.017$–0.024 on the probability scale) and candidates are strongly correlated (binary item correlation 0.57–0.68). The baseline prompt has 22.7%. 52 replay cells with 1000 replicates each; because replicates share the fixed pool, every uncertainty in this subsection is $\pm 1.96\times$ a delete-a-block SE (10 blocks of replicates) rather than a replicate SE or Wilson interval; it reflects the variation over search sets given the pool, not the uncertainty about the pool itself, which we do not estimate.

Gap. (Fig. 4.) The naive report is optimistic here too. For $K = 300$ and $n = 100$ the selected configuration is reported at 33.7% and scores 28.0% on fresh items: a 5.7-point gap (block SE 0.11). Against the baseline's 22.7%, the reported improvement is 11.0 points while the real one is 5.3: about twice as large. The gap is 8.7 points at $n = 50$ and 1.9 at $n = 500$ ($K = 300$), and 4.6 points for the 24 base prompts alone ($n = 100$). It grows with K and falls with n like the extreme-value formula evaluated with the matrix's own (σ, r, τ), which is within 0.51 points of every one of the 35 exhaustive cells (mean difference -0.04 points), although 13 of the 35 differ by more than 1.96 block SE, mostly at small K and $n = 50$ where it over-predicts (e.g. $K = 2$, $n = 50$: 1.80 predicted, 1.29 observed). The parameters are estimated from the same matrix, so this checks the functional form, not a held-out prediction. The synthetic grid of T0 gives a larger gap than the real one in 28 of its 30 nearest cells (median ratio 1.6; its correlation grid does not contain the matrix's), so we treat T0 as a model of the mechanism, not a calibrated replica (Fig. 6, appendix). Halving and greedy search are mildly less optimistic than exhaustive search (4.0 and 4.1 vs. 4.5 points at $n = 100$, $K = 24$; 5.2 vs. 5.7 at $K = 300$). The gap is present in every one of the 14 MMLU-Pro subjects (3.3 to 7.0 points at $n = 100$, $K = 24$; Fig. 7, appendix).

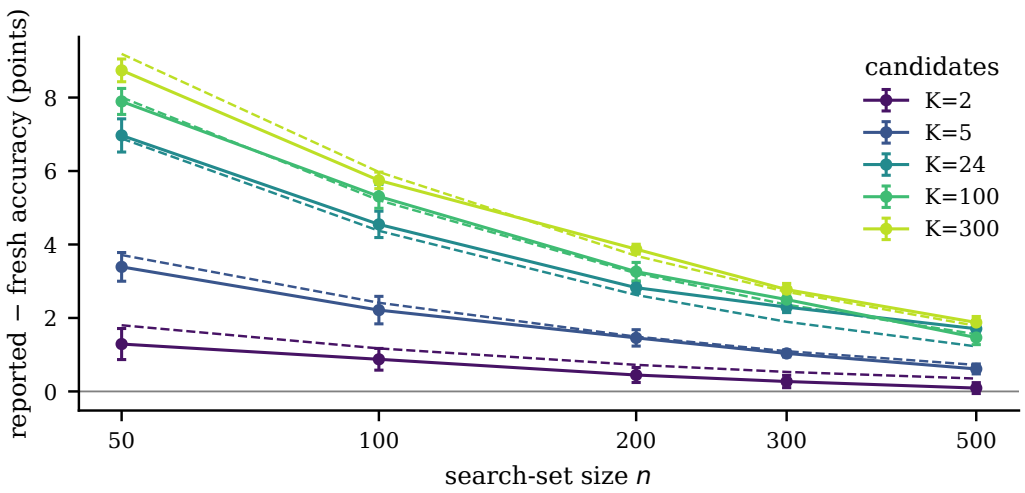


Figure 4: **Optimism of the naive report on the real outcome matrix (T1).** Reported minus fresh-item accuracy of the selected configuration against n, for K candidates searched exhaustively (Qwen2.5-1.5B-Instruct, MMLU-Pro; 1000 replicates per cell). Bars: mean $\pm 1.96\times$ delete-a-block SE (replicates share one 2000-item pool, so the replicate SE is too small); dashed: the closed form fed with the matrix's own (σ, r, τ).

Table 3: **Calibration of reporting methods on the real matrix (T1),** $K = 300$**,** $n = 100$**, 1000 replicates** ($n = 500$ in the appendix). Bias and RMSE in points against each method's own target; coverage ± 1.96 block SE; median width (points); items used for the report.

n	method	bias (pts)	RMSE (pts)	coverage $\pm$	width (pts)	items for report
100	M0	+5.75 $\pm$ 0.22	7.05	0.771 $\pm$ 0.032	18.3	100
100	M1_f0.5	-0.08 $\pm$ 0.31	6.21	0.954 $\pm$ 0.013	24.2	50
100	M2	-0.05 $\pm$ 0.23	4.95	0.923 $\pm$ 0.016	17.3	100
100	M4_simult	+5.75 $\pm$ 0.22	7.05	0.988 $\pm$ 0.005	29.5	100
100	M4_cond	-2.40 $\pm$ 1.46	20.77	0.944 $\pm$ 0.008	29.4	100
100	M4_hybrid	+1.11 $\pm$ 0.31	7.48	0.953 $\pm$ 0.008	24.1	100
100	M5_eb	-1.41 $\pm$ 0.19	4.05	0.109 $\pm$ 0.011	1.0	100
100	M5_ladder	-4.13 $\pm$ 0.40	8.04	–	–	100

Reporting methods and decisions. (Table 3.) The naive interval covers $\theta(\hat{c})$ in 77.1 $\pm$ 3.2% ($n = 100$) and 80.6 $\pm$ 2.2% ($n = 500$; ± 1.96 block SE) of replicates at $K = 300$ and is biased by $+5.7$ and $+1.9$ points. Hold-out (M1, 0.95/0.94 at half split), the hybrid conditional interval (0.953/0.935) and cross-fitting (0.92/0.88) are close to nominal and unbiased; the conditional interval is nominal but its point estimate is heavy-tailed (RMSE 20.8 points at $n = 100$); the simultaneous bound is conservative (0.988) and as biased as M0; the empirical-Bayes interval again fails (0.11/0.22). Bootstrap optimism correction removes most but not all of the bias ($+1.8$ points at $n = 100$, $K = 300$, vs. $+5.7$ for M0; appendix). Because the 276 ensembles are not independent systems and the candidates differ little, "tuned beats baseline" is almost always true here, so false-claim rates are not identifiable from these cells; we use a paired design instead (Table 4): two systems tuned with $K_A = 75$ and $K_B = 5$ candidates on the same $n = 100$ items, where the true difference is on average 0.5 points and A is truly better in only 61.7% of the replicates. Naive reports declare A better in 90.3% of replicates (the reported difference is $+3.0$ points against a true $+0.5$); among the 383 replicates in which A is not truly better, 87.5% still call it better with M0, versus 43–56% for hold-out, cross-fitting and the selection-aware point rules. With a rule that requires A's 95% lower bound to exceed B's estimate the false-winner rate is 0% for M0 and the simultaneous bound and 3.1–3.4% for hold-out, cross-fitting and the conditional/hybrid intervals, but those rules detect a truly better A in only 7–9% of replicates (M0: 1.8%); the equal-K pairs have no true difference, and every report order there is a coin flip (flip rates near 0.5 by construction).

Table 4: **Ranking two tuned systems on the same search set (T1).** A and B are tuned exhaustively from disjoint candidate subsets of K_A and K_B configurations; 1000 replicates per cell, ± 1.96 block SE. Flip: the reported order differs from the true order (a tie counts as a flip). The 75 vs. 5 pair has the largest search-effort contrast; the equal-K pairs are controls with no true difference.

K_A vs K_B	n	true $\theta_A - \theta_B$ (pts)	reported $\hat{\theta}_A - \hat{\theta}_B$ (pts)	P(A truly better)	P(claim A>B)	P(rank flip)
75 vs 5	100	+0.50 $\pm$ 0.10	+3.01 $\pm$ 0.10	0.62	0.90 $\pm$ 0.02	0.41 $\pm$ 0.03
75 vs 5	500	+0.98 $\pm$ 0.10	+1.67 $\pm$ 0.12	0.74	0.91 $\pm$ 0.02	0.25 $\pm$ 0.02
75 vs 75	100	+0.00 $\pm$ 0.06	+0.03 $\pm$ 0.14	0.47	0.37 $\pm$ 0.05	0.63 $\pm$ 0.04
75 vs 75	500	-0.05 $\pm$ 0.08	-0.00 $\pm$ 0.03	0.47	0.44 $\pm$ 0.02	0.52 $\pm$ 0.03
24 vs 24	100	+0.07 $\pm$ 0.08	+0.02 $\pm$ 0.07	0.49	0.37 $\pm$ 0.02	0.58 $\pm$ 0.04
24 vs 24	500	+0.07 $\pm$ 0.10	+0.02 $\pm$ 0.07	0.53	0.47 $\pm$ 0.04	0.42 $\pm$ 0.04
5 vs 5	100	-0.01 $\pm$ 0.11	+0.03 $\pm$ 0.14	0.49	0.42 $\pm$ 0.02	0.51 $\pm$ 0.02
5 vs 5	500	+0.06 $\pm$ 0.08	-0.01 $\pm$ 0.09	0.51	0.47 $\pm$ 0.04	0.38 $\pm$ 0.04

5 Discussion and Limitations

What to report. The evidence supports a conditional answer. The naive report of a searched system is biased upward whenever more than one configuration is compared on the same items; the bias is of the order of the true differences between configurations when n is a few hundred and K is large, so reported improvements of a few points are not interpretable without n and K. Reporting the score of $\hat{c}$ on items not used for the search (M1) removes the bias at the price of items; a conditional/hybrid selection-aware interval (M4) gives nominal coverage for exhaustive search without extra items but needs the full score table and, in our runs, is wide; cross-fitting (M2) estimates the performance of the procedure, with mildly undercovering intervals. At minimum a report should state K, n and the search procedure, and avoid comparing systems tuned with different K or n on reported scores. Methods that look attractive can fail: empirical-Bayes shrinkage of disclosed scores assumes independent noise, and the Ladder-style report depends on a free step size.

Limitations. (i) The real-LLM evidence is one 1.5B-parameter model on a pool of 2000 MMLU-Pro questions on CPU; no larger model, second benchmark or second task was run, so we make no statement about frontier systems. (ii) The candidates are prompt formats and two-prompt ensembles; their true spread is small and they are correlated, which is a low-signal regime; real prompt/program search with instruction proposals may have larger spread and adaptive proposals that our three procedures (exhaustive, halving, greedy) only approximate. (iii) T1 is a replay on cached outcomes, *not* a live run of DSPy/MIPRO; the live-tool tier of our plan was not done. (iv) The pool is fixed, and the uncertainty in T1 is over search sets given the pool; between-pool uncertainty was not estimated; the block SEs reflect variation over search sets given the pool, not the pool. (v) T0 is a model of the mechanism, not calibrated to T1 (its gap is larger than the real one in 28 of 30 nearest cells, median ratio 1.6). (vi) The extreme-value formula is heuristic. (vii) Selection-aware inference (M4) is implemented only for exhaustive search; for halving and greedy only M0–M3 and M5 are available. (viii) Coverage and decision rates in cells with few replicates in a regime (e.g. false-claim regimes at large τ) have wide intervals (Table 2). (ix) Compute (Table 8, appendix): the synthetic simulations took 3,807 s of summed job wall time, scoring 24 prompts on 2000 items (48,000 model calls) took 81,172 s elapsed on 8 parallel 2-vCPU CPU jobs, and the replays on cached outcomes 1,217 s; the 0.5B model arm, a second model and a live DSPy/MIPRO run were not done.

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A Additional tables

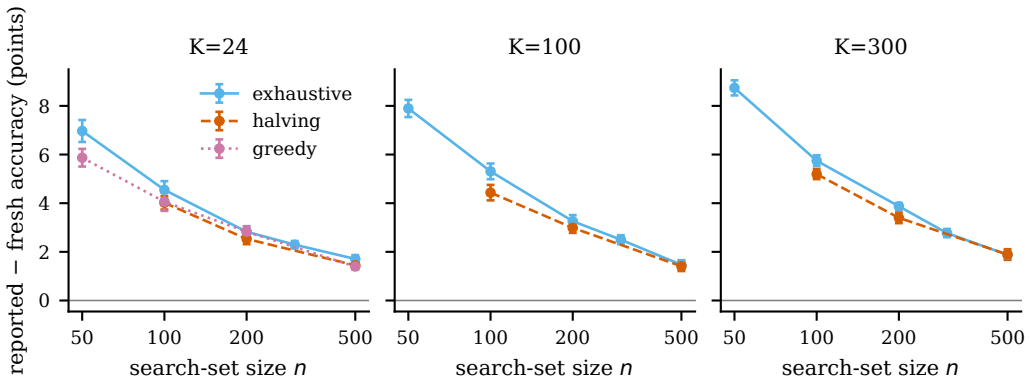


Figure 5: **Search procedure (T1).** Optimism by exhaustive, successive-halving and greedy search (greedy only at $K = 24$); panels $K = 24, 100, 300$; ± 1.96 block SE.

Table 5: **Full calibration table (T0)**, $\tau \in \{0, 0.1\}$, $K = 200$, $n \in \{100, 500\}$, 1000 replicates per cell; columns as Table 1.

τ	n	method	bias (pts)	RMSE (pts)	coverage [Wilson 95%]	width (pts)	items for report
0	100	M0	+9.85 ± 0.21	10.40	0.341 [0.312, 0.371]	18.9	100
0	100	M1_f0.2	+0.06 ± 0.32	5.14	0.962 [0.948, 0.972]	19.7	80
0	100	M1_f0.5	-0.08 ± 0.40	6.38	0.964 [0.951, 0.974]	24.6	50
0	100	M1_f0.8	+0.06 ± 0.65	10.49	0.967 [0.954, 0.976]	37.3	20
0	100	M2	-0.03 ± 0.34	5.49	0.880 [0.858, 0.899]	17.7	100
0	100	M4_simult	+9.85 ± 0.21	10.40	0.986 [0.977, 0.992]	34.2	100
0	100	M4_cond	-3.99 ± 1.86	30.22	0.961 [0.947, 0.971]	45.8	100
0	100	M4_hybrid	+2.61 ± 0.46	7.94	0.964 [0.951, 0.974]	27.3	100
0	100	M5_eb	-0.09 ± 0.17	2.81	0.208 [0.184, 0.234]	1.3	100
0	100	M5_ladder	-1.17 ± 0.63	10.26	–	–	100
0	100	M5_ladder_se	+8.24 ± 0.25	9.19	–	–	100
0	500	M0	+4.47 ± 0.09	4.70	0.414 [0.384, 0.445]	8.3	500
0	500	M1_f0.2	+0.03 ± 0.14	2.32	0.944 [0.928, 0.957]	8.9	400
0	500	M1_f0.5	+0.16 ± 0.19	3.00	0.950 [0.935, 0.962]	11.3	250
0	500	M1_f0.8	+0.25 ± 0.29	4.66	0.937 [0.920, 0.950]	17.7	100
0	500	M2	+0.08 ± 0.16	2.58	0.894 [0.873, 0.912]	8.0	500
0	500	M4_simult	+4.47 ± 0.09	4.70	0.977 [0.966, 0.985]	14.9	500
0	500	M4_cond	-2.01 ± 0.77	12.58	0.961 [0.947, 0.971]	21.5	500
0	500	M4_hybrid	+0.91 ± 0.20	3.40	0.962 [0.948, 0.972]	11.8	500
0	500	M5_eb	+0.02 ± 0.08	1.30	0.175 [0.153, 0.200]	0.6	500
0	500	M5_ladder	-1.68 ± 0.34	5.69	–	–	500
0	500	M5_ladder_se	+3.64 ± 0.11	4.03	–	–	500
0.1	100	M0	+7.40 ± 0.25	8.40	0.680 [0.650, 0.708]	19.1	100
0.1	100	M1_f0.2	-0.01 ± 0.32	5.13	0.956 [0.941, 0.967]	20.3	80
0.1	100	M1_f0.5	-0.29 ± 0.42	6.73	0.943 [0.927, 0.956]	25.4	50
0.1	100	M1_f0.8	+0.05 ± 0.67	10.78	0.953 [0.938, 0.964]	38.6	20
0.1	100	M2	-0.35 ± 0.37	5.90	0.886 [0.865, 0.904]	18.4	100
0.1	100	M4_simult	+7.40 ± 0.25	8.40	0.993 [0.986, 0.997]	34.6	100
0.1	100	M4_cond	-3.21 ± 1.44	23.42	0.970 [0.957, 0.979]	41.5	100
0.1	100	M4_hybrid	+1.34 ± 0.46	7.52	0.964 [0.951, 0.974]	27.5	100
0.1	100	M5_eb	-4.67 ± 0.24	6.05	0.278 [0.251, 0.307]	5.3	100
0.1	100	M5_ladder	-1.48 ± 0.63	10.20	–	–	100
0.1	100	M5_ladder_se	+6.07 ± 0.28	7.56	–	–	100
0.1	500	M0	+2.23 ± 0.13	3.04	0.827 [0.802, 0.849]	8.6	500
0.1	500	M1_f0.2	+0.08 ± 0.15	2.41	0.944 [0.928, 0.957]	9.4	400
0.1	500	M1_f0.5	+0.17 ± 0.20	3.21	0.929 [0.911, 0.943]	11.9	250
0.1	500	M1_f0.8	+0.22 ± 0.31	4.93	0.949 [0.934, 0.961]	18.7	100
0.1	500	M2	+0.16 ± 0.17	2.70	0.882 [0.861, 0.901]	8.5	500
0.1	500	M4_simult	+2.23 ± 0.13	3.04	0.999 [0.994, 1.000]	15.4	500
0.1	500	M4_cond	-1.14 ± 0.52	8.50	0.953 [0.938, 0.964]	14.5	500
0.1	500	M4_hybrid	+0.01 ± 0.21	3.46	0.961 [0.947, 0.971]	11.9	500
0.1	500	M5_eb	-1.12 ± 0.13	2.33	0.835 [0.811, 0.857]	6.6	500
0.1	500	M5_ladder	-1.99 ± 0.25	4.52	–	–	500
0.1	500	M5_ladder_se	+1.70 ± 0.14	2.84	–	–	500

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and Sec. 1 state the measured optimism, the decision consequences and the scope limits (one small model, replay not live search, no live tool run); Sec. 5 lists them.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.

Table 6: **Bootstrap optimism correction (M3) with the other methods (T0)**, $n = 200$, $K = 200$, 300 replicates per cell.

τ	n	method	bias (pts)	RMSE (pts)	coverage [Wilson 95%]	width (pts)	items for report
0	200	M0	+6.96 ± 0.26	7.33	0.383 [0.330, 0.439]	13.3	200
0	200	M1_f0.2	+0.29 ± 0.43	3.78	0.937 [0.903, 0.959]	14.1	160
0	200	M1_f0.5	-0.70 ± 0.53	4.70	0.943 [0.911, 0.964]	17.5	100
0	200	M1_f0.8	-0.67 ± 0.78	6.91	0.957 [0.927, 0.975]	27.4	40
0	200	M2	-0.46 ± 0.46	4.04	0.887 [0.846, 0.918]	12.5	200
0	200	M3_opt	+1.93 ± 0.28	3.16	–	–	200
0	200	M3_632	+2.54 ± 0.26	3.45	–	–	200
0	200	M4_simult	+6.96 ± 0.26	7.33	0.993 [0.976, 0.998]	24.0	200
0	200	M4_cond	-3.80 ± 2.34	21.03	0.973 [0.948, 0.986]	35.2	200
0	200	M4_hybrid	+1.34 ± 0.57	5.25	0.967 [0.940, 0.982]	19.1	200
0	200	M5_eb	+0.05 ± 0.22	1.95	0.183 [0.144, 0.231]	0.9	200
0	200	M5_ladder	+3.40 ± 0.41	4.96	–	–	200
0	200	M5_ladder_se	+5.59 ± 0.29	6.15	–	–	200
0.1	200	M0	+4.74 ± 0.37	5.75	0.703 [0.649, 0.752]	13.6	200
0.1	200	M1_f0.2	-0.12 ± 0.42	3.68	0.963 [0.936, 0.979]	14.6	160
0.1	200	M1_f0.5	+0.21 ± 0.57	5.02	0.940 [0.907, 0.962]	18.5	100
0.1	200	M1_f0.8	-0.42 ± 0.88	7.82	0.927 [0.891, 0.951]	28.4	40
0.1	200	M2	+0.00 ± 0.52	4.61	0.850 [0.805, 0.886]	13.2	200
0.1	200	M3_opt	+0.94 ± 0.40	3.67	–	–	200
0.1	200	M3_632	+1.27 ± 0.39	3.68	–	–	200
0.1	200	M4_simult	+4.74 ± 0.37	5.75	0.993 [0.976, 0.998]	24.5	200
0.1	200	M4_cond	-3.47 ± 2.19	19.59	0.953 [0.923, 0.972]	27.4	200
0.1	200	M4_hybrid	+0.47 ± 0.63	5.55	0.967 [0.940, 0.982]	19.2	200
0.1	200	M5_eb	-2.61 ± 0.36	4.12	0.633 [0.577, 0.686]	7.6	200
0.1	200	M5_ladder	-2.89 ± 0.38	4.44	–	–	200
0.1	200	M5_ladder_se	+3.71 ± 0.39	5.07	–	–	200

Table 7: **Claim "A beats B" from reported scores (T0).** False-winner and detection rates (%, Wilson 95%).

method	τ=0, A: K=200 vs B: K=5 false winner, point	same false winner, lower bound	τ=0.1, same pair false winner, point	τ=0.1, same pair A truly better, point	τ=0, A: n=50 vs B: n=500 (K=50) false winner, point
M0	86.7 [84.5, 88.7]	0.4 [0.2, 1.0]	73.1 [66.0, 79.2]	93.2 [91.3, 94.8]	93.2 [91.5, 94.6]
M1_f0.2	48.5 [45.4, 51.6]	0.2 [0.1, 0.7]	40.4 [33.3, 47.8]	66.6 [63.3, 69.7]	48.5 [45.4, 51.6]
M1_f0.5	46.7 [43.6, 49.8]	0.3 [0.1, 0.9]	46.2 [38.9, 53.7]	68.8 [65.5, 71.8]	46.3 [43.2, 49.4]
M1_f0.8	47.8 [44.7, 50.9]	0.6 [0.3, 1.3]	36.8 [30.0, 44.3]	61.8 [58.4, 65.0]	47.9 [44.8, 51.0]
M2	47.8 [44.7, 50.9]	2.0 [1.3, 3.1]	35.7 [28.9, 43.1]	73.0 [69.9, 75.9]	48.0 [44.9, 51.1]
M4_simult	86.7 [84.5, 88.7]	0.0 [0.0, 0.4]	73.1 [66.0, 79.2]	93.2 [91.3, 94.8]	93.2 [91.5, 94.6]
M4_cond	54.4 [51.3, 57.5]	0.4 [0.2, 1.0]	39.8 [32.7, 47.2]	74.7 [71.6, 77.5]	63.2 [60.2, 66.1]
M4_hybrid	55.0 [51.9, 58.1]	0.3 [0.1, 0.9]	39.2 [32.2, 46.7]	74.2 [71.1, 77.0]	61.0 [57.9, 64.0]
M5_eb	46.4 [43.3, 49.5]	13.2 [11.2, 15.4]	41.5 [34.4, 49.0]	75.8 [72.7, 78.5]	47.5 [44.4, 50.6]
M5_ladder	8.1 [6.6, 10.0]	–	5.8 [3.2, 10.4]	13.6 [11.5, 16.1]	77.6 [74.9, 80.1]

- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Sec. 5 (Limitations i–ix) covers model scale, candidate family, replay versus live search, fixed pool, T0 calibration, heuristic formula, M4 only for exhaustive search, and compute.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors

Table 8: **Compute actually used.** Wall seconds per job on 2-vCPU containers of a shared host; the total is a sum of job times, not elapsed time. The scoring job's 81,172 s is first-to-last cached file of 8 parallel jobs; its tokens were not logged. The totals are job accounting: the separate leakage job (`t0leak`, 16 cells, 16,000 replicates) repeats cells that are also in the main grid, so the distinct synthetic grid is 216 cells and 203,400 replicates.

job	wall (s)	cells	replicates	LLM calls	LLM tokens
t0/shard0.cost.csv	742	54	51,200	0	0
t0/shard1.cost.csv	1,172	54	51,200	0	0
t0/shard2.cost.csv	695	54	50,500	0	0
t0/shard3.cost.csv	1,030	54	50,500	0	0
t0leak/shard0.cost.csv	168	16	16,000	0	0
score.py	81,172	24	2,000	48,000	–
t1.py shards 0-3	801	52	52,000	0	0
t1_pairs.py n=50/100/200/500	416	16	16,000	0	0
total	86,196	324	289,400	48,000	–

Table 9: **Calibration on the real matrix (T1),** $K = 300$**,** $n \in \{100, 500\}$; columns as Table 3. Uncertainty is ± 1.96 block SE; replicates share one pool (Limitation iv).

n	method	bias (pts)	RMSE (pts)	coverage $\pm$	width (pts)	items for report
100	M0	+5.75 $\pm$ 0.22	7.05	0.771 $\pm$ 0.032	18.3	100
100	M1_f0.2	-0.17 $\pm$ 0.38	4.97	0.946 $\pm$ 0.013	19.2	80
100	M1_f0.5	-0.08 $\pm$ 0.31	6.21	0.954 $\pm$ 0.013	24.2	50
100	M1_f0.8	-0.37 $\pm$ 0.65	10.23	0.955 $\pm$ 0.016	35.7	20
100	M2	-0.05 $\pm$ 0.23	4.95	0.923 $\pm$ 0.016	17.3	100
100	M3_opt	+1.76 $\pm$ 0.35	4.83	–	–	100
100	M3_632	+2.06 $\pm$ 0.34	4.84	–	–	100
100	M4_simult	+5.75 $\pm$ 0.22	7.05	0.988 $\pm$ 0.005	29.5	100
100	M4_cond	-2.40 $\pm$ 1.46	20.77	0.944 $\pm$ 0.008	29.4	100
100	M4_hybrid	+1.11 $\pm$ 0.31	7.48	0.953 $\pm$ 0.008	24.1	100
100	M5_eb	-1.41 $\pm$ 0.19	4.05	0.109 $\pm$ 0.011	1.0	100
100	M5_ladder	-4.13 $\pm$ 0.40	8.04	–	–	100
100	M5_ladder_se	+3.53 $\pm$ 0.27	5.57	–	–	100
500	M0	+1.88 $\pm$ 0.16	2.99	0.806 $\pm$ 0.022	8.1	500
500	M1_f0.2	+0.03 $\pm$ 0.16	2.60	0.901 $\pm$ 0.023	8.8	400
500	M1_f0.5	-0.09 $\pm$ 0.19	2.98	0.939 $\pm$ 0.019	11.1	250
500	M1_f0.8	-0.30 $\pm$ 0.29	4.70	0.931 $\pm$ 0.015	17.5	100
500	M2	-0.05 $\pm$ 0.21	2.56	0.875 $\pm$ 0.025	7.9	500
500	M3_opt	+0.20 $\pm$ 0.13	2.46	–	–	500
500	M3_632	+0.32 $\pm$ 0.13	2.44	–	–	500
500	M4_simult	+1.88 $\pm$ 0.16	2.99	0.976 $\pm$ 0.003	13.1	500
500	M4_cond	-1.55 $\pm$ 1.15	10.81	0.934 $\pm$ 0.014	12.1	500
500	M4_hybrid	+0.25 $\pm$ 0.15	3.20	0.935 $\pm$ 0.012	10.6	500
500	M5_eb	-1.94 $\pm$ 0.17	3.05	0.221 $\pm$ 0.028	0.5	500
500	M5_ladder	-3.59 $\pm$ 0.23	4.11	–	–	500
500	M5_ladder_se	+0.93 $\pm$ 0.14	2.63	–	–	500

should reflect on how these assumptions might be violated in practice and what the implications would be.

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Table 10: **False-winner and detection rates (%, ± 1.96 block SE) for "A beats B" on the real matrix (T1),** $n = 100$**.** "point": estimate of A exceeds that of B; "lower bd": A's lower bound exceeds B's estimate. Tuned-versus-baseline decisions are omitted: the baseline is almost always truly worse, so the false-claim rate is not identifiable.

method	75 vs 5 false winner, point	75 vs 5 false winner, lower bd	75 vs 75 false winner, point	75 vs 75 false winner, lower bd	75 vs 5 A truly better, point	75 vs 5 A truly better, lower bd
M0	87.5 ± 4.3	0.0 ± 0.0	35.4 ± 5.9	0.0 ± 0.0	92.1 ± 1.2	1.8 ± 0.8
M1_f0.5	43.6 ± 5.7	3.1 ± 1.2	43.2 ± 4.0	6.9 ± 2.5	50.1 ± 4.5	7.5 ± 1.4
M2	42.6 ± 7.3	3.1 ± 1.2	45.0 ± 3.9	2.3 ± 1.1	52.2 ± 5.2	6.3 ± 1.6
M4_simult	87.5 ± 4.3	0.0 ± 0.0	35.4 ± 5.9	0.0 ± 0.0	92.1 ± 1.2	0.0 ± 0.0
M4_cond	56.1 ± 6.7	3.4 ± 2.0	49.3 ± 6.1	9.7 ± 3.3	68.7 ± 2.7	6.6 ± 1.7
M4_hybrid	54.8 ± 6.7	3.1 ± 1.4	48.8 ± 5.1	7.4 ± 2.5	68.4 ± 3.5	8.6 ± 1.8
M5_eb	40.5 ± 3.8	14.9 ± 3.3	48.2 ± 3.6	2.1 ± 1.0	52.0 ± 2.9	24.6 ± 3.5
M5_ladder	7.3 ± 2.6	–	6.5 ± 2.4	–	7.6 ± 1.7	–

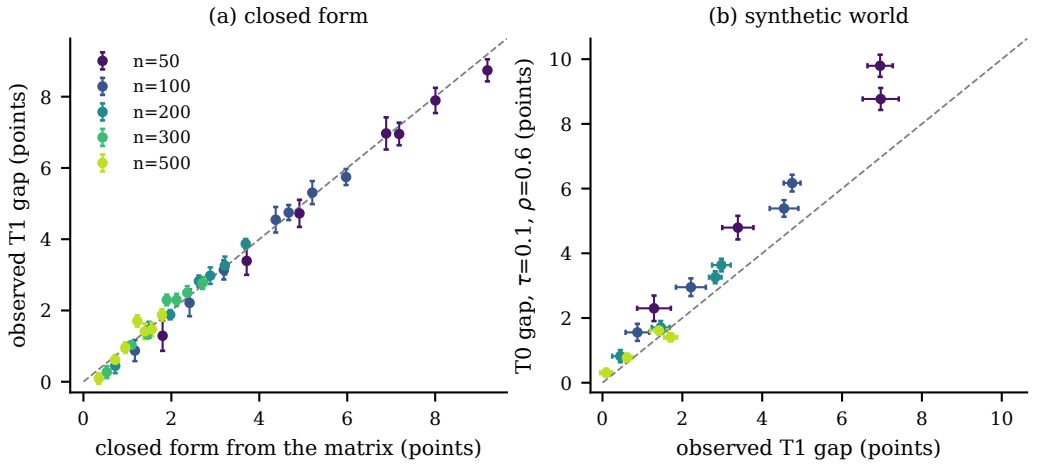


Figure 6: **Checks of the extreme-value form and of the synthetic world against the real matrix (T1).** (a) Observed gap against the closed form fed with the matrix's (σ, r, τ), all 35 exhaustive cells. (b) Observed T1 gap against the T0 cells with the same (n, K) at $\tau = 0.1$, $\rho = 0.6$ and baseline accuracy 0.3 (T0 is not calibrated to the matrix); dashed: $y = x$.

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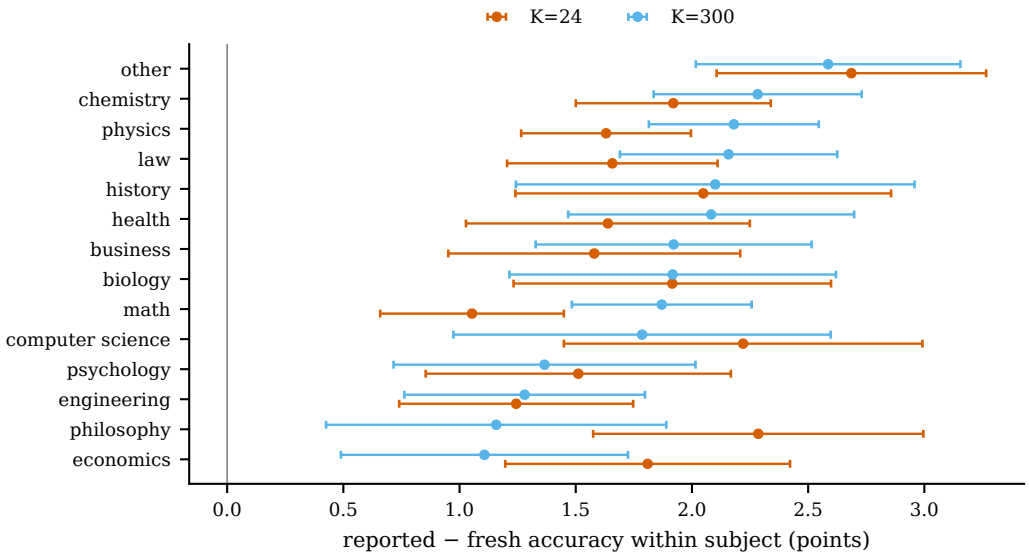


Figure 7: **Gap by MMLU-Pro subject (T1)**, $n = 500$, $K = 24$ and 300. A subject holds 16–63 of the 500 search items and replicates share items, so the bars understate uncertainty; the figure shows that optimism is present in every subject, not differences between subjects.

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